

Simple Alarm Clock INCOSE Systems Engineering Document

1. Business or Mission Analysis

1.1 Problem Statement

Users need a reliable, standalone device to wake them up at a specified time to maintain their daily schedules without relying exclusively on complex or distraction-prone devices like smartphones.

1.2 Mission Objectives

- Provide a clear, readable display of the current time.
- Allow users to easily set a specific alarm time.
- Sound a noticeable auditory alert when the current time matches the alarm time.

1.3 Key Stakeholders (Business Level)

- Primary Users (People waking up)
- Manufacturers

2. Stakeholder Needs and Requirements Definition

2.1 Use Cases / Operational Scenarios

- **UC1: Check Time** - User looks at the device to see the current time.
- **UC2: Set Alarm** - User inputs a specific time into the device to trigger an alert.
- **UC3: Wake Up** - The device sounds an alarm at the set time, waking the user. User disables the alarm.

2.2 Measures of Effectiveness (MOEs)

- Time accuracy (drift per month).
- Alarm volume (decibels at 1 meter).
- Display visibility (viewing angle and brightness).

3. System Requirements Definition

3.1 Functional Requirements

- The system shall display the current time.
- The system shall allow the user to set an alarm time.
- The system shall sound an alarm when the current time matches the alarm time.
- The system shall provide a mechanism to turn off the active alarm.

3.2 Non-Functional / Quality Requirements

- The alarm volume shall be at least 70 dB.
- The time display shall be visible in a dark room.

3.3 System Constraints

- The system shall operate on standard 120V AC wall power.